

This presentation is licensed under CC-BY-ND-NC 4.0

MeDaX: graph-based exploration of clinical data

Judith AH Wodke

MeDaX junior group, Medical Informatics Laboratory, University Medicine Greifswald

Heidelberg, 17.06.2026

Who am I?

- Studies of bioinformatics (B.Sc./M.Sc., Free University Berlin)
- PhD in theoretical biophysics (Humboldt Universität zu Berlin / Centro de Recerca Genetica Barcelona)
- Postdoc in theoretical biophysik (Humboldt Universität zu Berlin)
- Junior group leader in medical informatics (University Medicine Greifswald)

Who am I?

- Studies of bioinformatics (B.Sc./M.Sc., Free University Berlin)
- PhD in theoretical biophysics (Humboldt Universität zu Berlin / Centro de Recerca Genetica Barcelona)
- Postdoc in theoretical biophysik (Humboldt Universität zu Berlin)
- Junior group leader in medical informatics (University Medicine Greifswald)

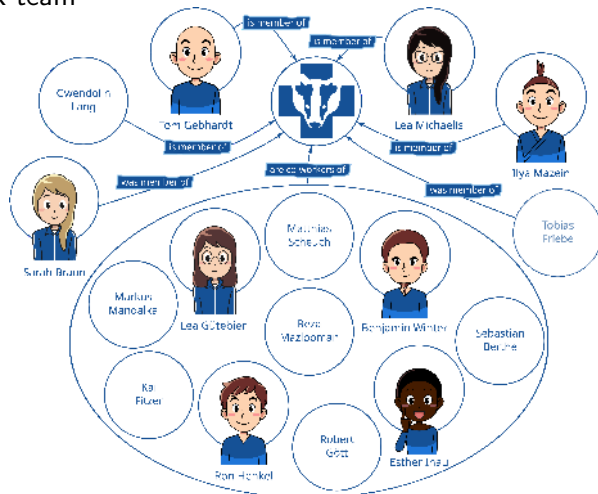
- **Research Interests:**

- Data management
- Knowledge graphs
- Mathematical modelling
- Standards
- Open science



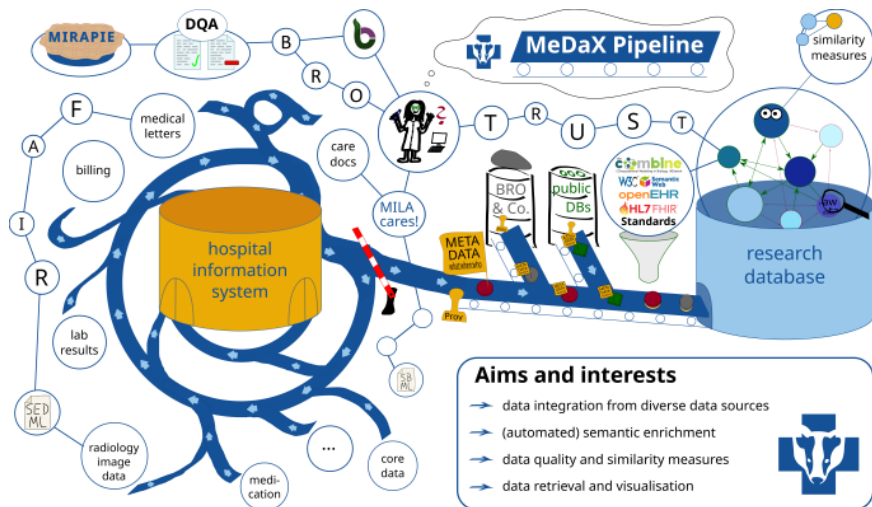
Who is MeDaX?

MeDaX team



Personalisierte Avatare von Tom Gebhardt, Originale von oksmith auf openclipart.org

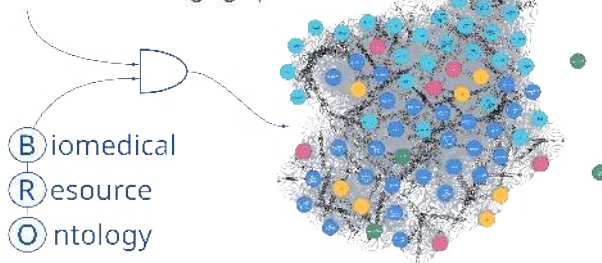
Goals and motivation



Why BioCypher?

- Right time, right place: I met Sebastian and he showed me code :)
- The BioCypher framework fulfills several requirements for the MeDaX project:
 - modular approach for integrating data from different sources
 - semantic enrichment supported
 - no direct dependencies from proprietary software
- I joined the BioCypher community, proposing MeDaX as a use case and supporting with improving FAIRness of the tooling
- And then we adopted the Biomedical Resource Ontology (BRO) to save time with not setting up our own ontology...

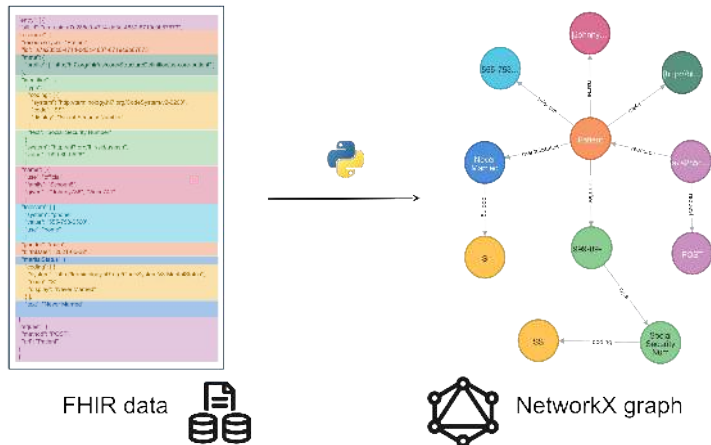
BioCypher + BRO = meta-graph



- Meta-graph repo: <https://github.com/biocypher/meta-graph>

credits: Sebastian Lobentanzer, -TEAM, and Julio Saez-Rodriguez

FHIR: Fast Healthcare Interoperability Resource



credits: Ilya Mazein

Generic graph structure



157 Resource types



Generic conversion



Large graphs

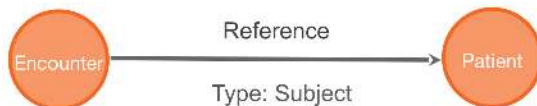
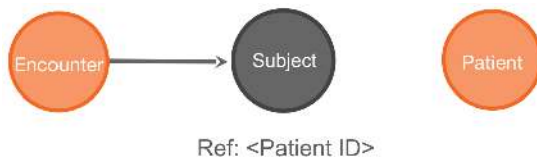


Optimisation



credits: Ilya Mazein

Reference path reduction



credits: Ilya Mazein

Feature path convolution



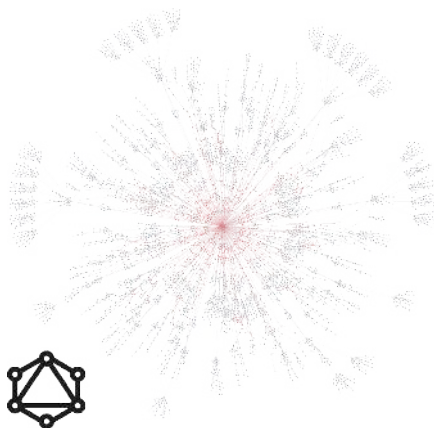
Feature paths before convolution



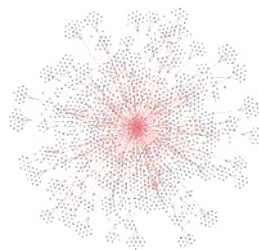
Feature paths after convolution

credits: Ilya Mazein

Structure optimisation



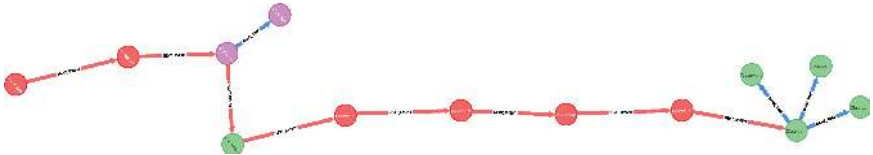
Generic bundle conversion



Optimised structure

credits: Ilya Mazein

"Patient Journey" example



credits: Ilya Mazein

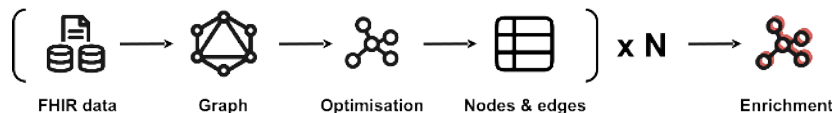
Prototype set up @ UMG DIZ

Prototype set up at



Challenge: memory shortage

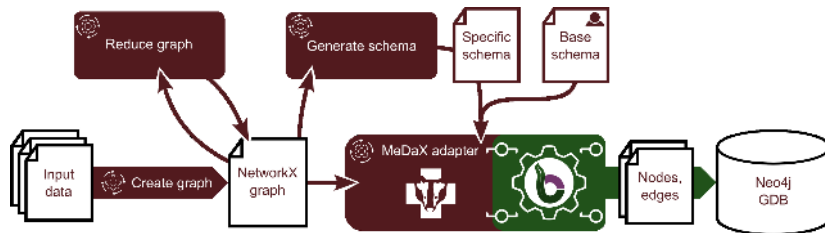
Solution: batching approach



tinyurl.com/medax-pipeline

credits: Ilya Mazein, Tom Gebhardt

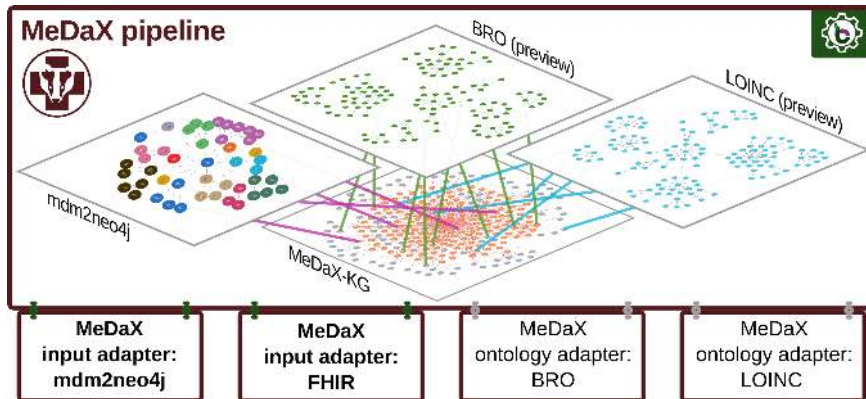
Automated KG generation



- generic NetworkX graph
- manually curated graph schema (Base schema)
- automatic schema update to prevent loss of information
- schema update now incorporated into BioCypher Core

credits: Tom Gebhardt, Ilya Mazein, Lea Michaelis, Sebastian Lobentanzer

Diverse data sources: mdm2neo4j graph integrated



credits: Lea Gütebier, Lea Michaelis, Ilya Mazein, Tom Gebhardt

- Minimal community standard for provenance documentation in biomedicine: MIRAPIE (MInimal Requirements for Automated Provenance Information Enrichment), including data model, ontology, and guideline
 - open repository: <https://codeberg.org/MIRAPIE>



- FAIR assessment of Medical Informatics Initiative core data set (CDS) modules *Person* and *Medication*: Guideline for FAIR (FHIR) metadata specifications
 - going to be published independently from the journal publication for easy reuse

credits: MIRAPIE community, Lea Michaelis

MEDIZIN
INFORMATIK
INITIATIVE

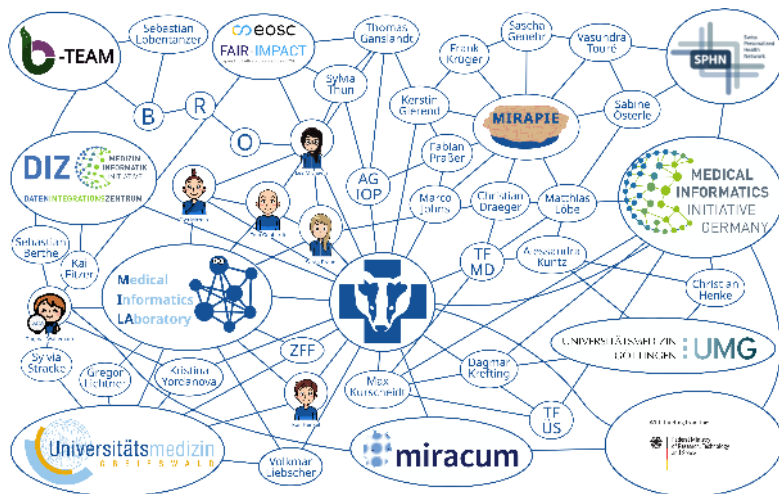


BRO - reduced to its core

- we started from 488 classes, added 1, deleted 123, and then
- manually curated 366 classes and definitions
- we now have 88 classes (35 of which have been newly defined)
- stringent definitions (any examples moved to comments)
- comprehensive linking to other ontologies
- about to be incremented to v5.0.0
- final testing and pull request pending
- manuscript in preparation
- it did not save time, but we learned a lot :)

- MeDaX Pipeline improvements, including
 - schema update
 - user interface
 - strukture-based search
 - visualisation of "patient journeys"
 - integration of terminologies/ontologies
- Data usage application: positive ethics vote and approval from the data protection officer in Greifswald (Rostock pending)
 - charakterisation of the full cohort
 - "proof of concept" for MeDaX pipeline
 - similarity and quality scores

Acknowledgments



Thanks for your attention!

References and Repositories

- BRO paper: Tenenbaum et al., J Biomed Inform 2011, 44(1):137-45, doi: 10.1016/j.jbi.2010.10.003
 - BRO repository: <https://github.com/biocypher/biomedical-resource-ontology>
 - FAIR paper: Wilkinson et al., Sci Data 2016, 3:160018, doi: 10.1038/sdata.2016.18
 - MeDaX paper: Mazein et al., doi: 10.3233/SHTI240423
 - MeDaX release: Gebhardt et al., <https://zenodo.org/records/15229077>
 - MIRAPIE paper: Wodke et al., doi: 10.2139/ssrn.5327042
 - MIRAPIE repository: <https://codeberg.org/MIRAPIE>
 - MIRAPIE ontology: Genehr et al., doi: 10.5281/zenodo.15608383
 - MIRAPIE guideline: Michaelis et al., doi: 10.5281/zenodo.17939689
 - BRAinS Graph: Gütebier L, et al. German Medical Science 2024; doi: 10.3205/24gmds015
- MILA homepage: <https://www.unimedizin-greifswald.de/medizininformatik/>